

Computer Science and Engineering Department, SVNIT, Surat
M.Tech. I – Semester I
Design and Analysis of Algorithm (CSDS103)
Lab Assignment 2

General Instructions

1. Implement the specified algorithms and measure their execution time for different input sizes.
2. Do not use library functions that directly implement the algorithm being studied.
3. Record the experimental results in tabular form and generate appropriate plots.
4. Compare the experimental results with the theoretical time complexity and comment on the observed behaviour.

Problem 1 – Experimental Analysis of Searching Algorithms

Implement the following searching algorithms:

- (a) Linear Search
- (b) Binary Search

Generate integer arrays of different sizes and perform searches for suitable elements. Consider the following different input sizes: 10, 50, 100, 200. Give the input using file.

Consider suitable cases such as the element being present near the beginning, middle, and end of the array, as well as the element being absent. Generate plots of **Input Size vs Execution Time** and compare the observed results with the theoretical time complexity of both algorithms.

Problem 2 – Experimental Analysis of Sorting Algorithms

Implement the following sorting algorithms:

- (a) Bubble Sort
- (b) Selection Sort
- (c) Insertion Sort

Generate arrays of different sizes and measure the execution time required by each algorithm. For each algorithm, consider the following types of input:

- Randomly ordered input
- Already sorted input
- Reverse sorted input

Use multiple input sizes and record the execution time for every case. Generate appropriate plots of **Input Size vs Execution Time** and compare the experimental results with the theoretical best-case and worst-case complexities.

Problem 3 – Experimental Analysis of Matrix Algorithms

Implement the following matrix operations:

- (a) Matrix Addition
- (b) Matrix Multiplication
- (c) Matrix Transpose

Perform the operations on square matrices of different sizes, for example,

$$10 \times 10, 50 \times 50, 100 \times 100, 200 \times 200, 500 \times 500$$

Generate the matrices programmatically and measure the execution time for each operation. Record the matrix size and corresponding execution time in tabular form.

Generate plots of **Matrix Size vs Execution Time** for each operation and compare the experimental results with their theoretical time complexities.